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Network-Validated Distributed ADMM Coordination of Cold-Climate Electric Heating under Communication Failures

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NOMENCLATURE

N	Number of heating agents (homes), $N = 74$.
T	Number of time steps over the day, $T = 96$.
$h_{i,t}$	Baseline (thermostat-driven) heating power of home i at step t.
$x_{i,t}$	Coordinated heating power of home i at step t.
$\hat{x}_{i,t}$	Imputed heating power of a non-responsive home i.
B_t	Fixed (non-deferrable) background aggregate load at step t.
c	Constant flat-aggregate target level (energy-conserving).
α	Per-step comfort deferrability band, $\alpha = 0.5$.
λ	Comfort-deviation weight, $\lambda = 0.15$.
μ	Aggregate-flattening weight, $\mu = 1.0$.
ρ_{ADMM}	ADMM penalty parameter, $\rho_{\mathrm{ADMM}} = 1.0$.
ϱ	Non-responsive (communication-failure) fraction of agents.
z	ADMM auxiliary shared-aggregate variable.
u	ADMM scaled dual variable.
ν_i	Scalar dual of the daily-energy hyperplane in the projection.
$V_{\min}$	Worst-of-day minimum bus voltage (p.u.).

I. INTRODUCTION

THE electrification of heating in cold-climate regions concentrates distribution-system stress into a few winter hours. In the province of Québec, where electric space heating is the dominant heating technology, the system peak is driven almost entirely by simultaneous resistive and heat-pump heating on the coldest mornings and evenings. As distributed energy resources (DERs) proliferate and heating loads grow, these synchronized peaks threaten thermal limits on feeder trunk conductors and depress voltages far from the substation. Because the flexibility lives behind thousands of meters and not at a single controllable node, centralized dispatch scales poorly; decentralized coordination, in which each home solves a small local problem and exchanges only an aggregate signal, is the natural fit.

The alternating direction method of multipliers (ADMM) [1] is a workhorse for this setting: its sharing form lets a coordinator flatten a common resource (here, total feeder load) while each agent retains private control of its own heating schedule. ADMM-based demand response is well studied, yet two gaps undermine its real-world credibility.

The first gap is fragility to communication failures. Multi-agent coordination implicitly assumes every agent reports on every iteration; when a fraction of agents fall silent, the aggregate the coordinator optimizes is no longer the aggregate that materializes on the wire. How badly the coordinated outcome degrades — and how an operator might recover it by forecasting the missing contributions — is rarely quantified end to end.

The second gap is physical validity. The literature typically stops at the coordinated aggregate kilowatt signal and reports peak or peak-to-average reductions. It almost never asks the operational question that matters: is the coordinated aggregate actually feasible on the feeder? A flatter aggregate is worthless if it still overloads a trunk line or sags a remote bus below limits.

This paper closes the loop between coordination and feeder physics. We adapt the sharing-ADMM-plus-imputation methodology from a prior multi-agent thesis and extend it with an AC power-flow validation stage, turning a purely signal-level study into a network-validated one. Our contributions are:

- A sharing-ADMM coordinator for deferrable electric heating that flattens the feeder aggregate subject to per-

home comfort bounds and a daily-energy conservation constraint (load is shifted, never curtailed).
- A LightGBM imputer that reconstructs the heating of non-responsive agents from ambient temperature, time-of-day, and nameplate features, letting the coordinator stay accurate as the non-responsive fraction grows, together with a comparison of four learned estimators against naive and no-imputation baselines under a realized-load model that isolates the value of imputation.
- An AC power-flow validation stage on a synthetic LV residential feeder of 74 literal all-electric homes behind a single 400 kVA MV/LV distribution transformer in pandapower, in which the homes load the transformer directly and one-for-one (no artificial scaling), and which records worst-of-day transformer thermal loading and minimum bus voltage — closing the loop on physical feasibility.
- A fully reproducible study — deterministic, free of proprietary data, and built on an open-source toolchain — in which every reported number is generated from the run rather than transcribed by hand.

The central finding is that coordination not only flattens the aggregate (a 9.8% peak cut) but clears a 109.6% transformer thermal overload to 98.5%, and that this cleared state holds under communication failure up to a non-responsive fraction of about 0.70 before re-violating near 0.75. Throughout, we keep the communication-failure fraction (denoted ϱ) notationally distinct from the ADMM penalty (ρ_{ADMM}) so the two roles of the symbol are never conflated; the Nomenclature collects all symbols. We also stress two honest caveats that the results section quantifies: the flattening preserves per-home comfort by construction (heat is re-timed, never withheld), and because the objective targets a flat profile rather than the time-of-use price, the energy-bill saving is deliberately modest — the value of this demand response is capacity relief, demonstrated here as restored feeder feasibility, not a cheaper bill.

The remainder of the paper is organized as follows. Section IV is preceded by the related work and the system model; Section II surveys ADMM-based demand response, multi-agent resilience, telemetry imputation, and distribution power flow, and states our delta against each. Section III formalizes the agents, the feeder, and the communication-failure model. Section IV develops the sharing-ADMM updates, the LightGBM imputer, and the power-flow validation loop. Section V details the case-study configuration, Section VI reports the aggregate, robustness, and network-feasibility results, and Section VII concludes.

II. RELATED WORK

Our contribution sits at the intersection of five strands of literature: distributed (ADMM/consensus) coordination, network-aware demand response, resilience of distributed coordination to communication failures, machine-learning imputation of missing load telemetry, and the flexibility of residential electric heating. We review each and then make the gap explicit (Table 1).

A. ADMM AND CONSENSUS FOR DISTRIBUTED ENERGY COORDINATION

ADMM [1] decomposes a coupled optimization into local subproblems linked by a consensus or sharing constraint, and is now a standard tool for distributed energy coordination. The sharing form (Boyd et al., Section 7.3) is particularly suited to demand flattening, where many agents contribute to one shared resource. It underpins distributed optimal power flow [2], communication-efficient and consensus-based demand response [3], [4], asynchronous microgrid energy management [5], and hierarchical transactive coordination of home energy management systems [6]. These works establish that ADMM converges to a flattened or cost-optimal aggregate; with few exceptions they treat the aggregate signal as the deliverable and do not verify that it is feasible on the underlying feeder.

B. NETWORK-AWARE COORDINATION

A second strand embeds network physics in the coordination itself. Distributed AC optimal power flow couples ADMM with the branch-flow equations [7], and recent work coordinates distributed energy resources with inverter Volt-VAr control through a three-block AC-OPF ADMM [8]. Network-constrained demand-response pricing [9], ADMM-based congestion-management markets [10], and capacity-constrained electric-vehicle charging [11] all respect feeder limits during coordination. A parallel line handles uncertainty through chance-constrained AC-OPF, enforcing probabilistic voltage and thermal limits for renewables [12] and flexible-load ensembles [13]. This literature proves that network-feasible coordination is achievable, but each work assumes perfect, complete communication: every agent reports on every iteration.

C. RESILIENCE TO COMMUNICATION FAILURES

A third strand studies how coordination degrades when agents fall silent. Brown et al. show that even a single lost link can drive multi-agent systems to arbitrarily poor states [14], and fault-tolerant consensus is a mature survey topic [15]. In distributed optimization, stochastic communication delay slows ADMM-based OPF [16], lossy links are handled by robust dispatch [17] and relaxed asynchronous ADMM with convergence guarantees [18], and adversarial or non-responsive agents can be detected and isolated to preserve convergence [19]. Learning has begun to enter the loop — e.g. to warm-start consensus-ADMM OPF [20]. This literature characterizes when coordination breaks and offers robust-by-design or detect-and-isolate responses, but it rarely reconstructs a silent agent's contribution, and never propagates that reconstruction error into feeder physics.

D. IMPUTATION AND FORECASTING OF MISSING TELEMETRY

When an agent falls silent the coordinator must estimate its consumption. Machine-learning imputation of missing smart-meter data spans denoising diffusion models [21], correlation-clustering for faulted grid data [22], and low-rank recovery of asynchronous meter streams [23]. For the closely related task of short-term load prediction, gradient-boosted trees are competition-winning regressors [24], [25], motivating our use of LightGBM [26] to impute a silent home's heating from weather and calendar features. Crucially, in all of this work the imputer or forecaster is evaluated in isolation — by held-out RMSE — and is never embedded in a distributed coordinator nor propagated into a power-flow feasibility check.

E. RESIDENTIAL ELECTRIC-HEATING FLEXIBILITY

Finally, the resource we coordinate — thermostatically controlled electric heating — is itself well studied. Aggregate thermostatic flexibility can be quantified for demand response [27], direct electric space heating combined with partial thermal storage shifts load under comfort constraints via linear programming [28], heat-pump-plus-storage flexibility reduces peak demand and generation capacity [29], and field experiments confirm the aggregate effect and its rebound dynamics [30]. In cold climates specifically, electrically heated buildings with thermal storage enable peak shaving [31], and multi-unit residential flexibility and predictive heating control have been demonstrated in Québec-style winter-peaking contexts [32], [33]. The distributed-control variants closest to ours coordinate large TCL populations through asynchronous greedy [34] or resilient-consensus [35] schemes — yet these remain abstract control formulations without an AC feeder model, and the broader thermal-DR literature is overwhelmingly centralized and not network-validated.

F. THE GAP

Table 1 positions representative prior work along the five capabilities our study requires together: distributed ADMM/consensus coordination, AC power-flow validation, an explicit communication-failure model, machine-learning imputation of the missing agents, and uncertainty quantification of the resulting risk. Every ingredient exists — but only in disjoint subsets. Distributed-ADMM coordinators assume perfect communication; network-aware and chance-constrained methods are centralized or assume complete reporting; resilience methods detect or tolerate failures rather than reconstruct the missing agents; imputation accuracy is studied standalone; and thermal-DR studies seldom touch the feeder at all. To our knowledge no prior work closes the full loop — distributed ADMM coordination of cold-climate electric heating, machine-learning imputation of communication-failed homes, AC power-flow validation on a distribution feeder, and Monte-Carlo quantification of the resulting violation probability under a swept non-responsive fraction. That closed loop is the contribution of this paper.

III. SYSTEM MODEL

A. AGENTS AND DEFERRABLE HEATING

We model $N = 74$ representative cold-climate electric-heating homes over a single synthetic Trois-Rivières cold day at 15-minute resolution ($T = 96$ time steps). Each home i has a thermostat-driven baseline heating profile $h_{i,t}$ produced by a behavioral model conditioned on the ambient temperature, plus a fixed (non-deferrable) background load. The coldest-day minimum temperature is $-21.9\,^\circ\mathrm{C}$ (daily mean $-18.3\,^\circ\mathrm{C}$), which sets a demanding heating duty.

Heating is treated as deferrable within a comfort band of $\pm 50\%$ of the baseline ($\alpha = 0.5$): the coordinator may shift when a home heats but must conserve the home's total daily heating energy, so demand is shifted rather than curtailed. With $x_{i,t}$ the coordinated heating power of home i at step t, each agent is constrained by

$$(1-\alpha)\,h_{i,t} \le x_{i,t} \le (1+\alpha)\,h_{i,t}, \qquad \sum_t x_{i,t} = \sum_t h_{i,t}. \tag{1}$$

The box bound preserves thermal comfort at every step, and the equality plane preserves daily heating energy. Background load is held fixed and is not subject to (1).

B. THE FEEDER

Physical feasibility is assessed on a synthetic low-voltage (LV) residential feeder modeled in pandapower [36]. The 74 all-electric homes are connected as literal loads behind a single 400 kVA MV/LV distribution transformer (25 kV / 0.4 kV) and split across four parallel LV feeders. There is no artificial scaling: the homes are modeled one-for-one and their aggregate heating loads the transformer directly. The binding constraint is therefore the transformer's thermal loading — the classic residential winter-peak, cold-load-pickup limit — with the uncoordinated coincident peak of 401.6 kW driving the 400 kVA transformer to 109.6% loading. LV bus voltage is a secondary, healthy severity measure. All loads use a power factor of 0.95.

C. COMMUNICATION-FAILURE MODEL

At coordination time a fraction ϱ of the agents become non-responsive and report nothing to the coordinator. We use ϱ for this communication-failure fraction throughout, kept notationally distinct from the ADMM penalty ρ_{ADMM} of Section IV. We sweep $\varrho \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$, corresponding to 0, 15, 30, 44, and 59 silent homes out of 74. For each non-responsive agent the coordinator substitutes an imputed heating profile $\hat{x}_{i,t}$ produced by a trained regressor (Section IV) and proceeds as if it were the true contribution. The imputation error therefore propagates first into the coordinated aggregate and ultimately into the feeder power flow, which is exactly the propagation path this study quantifies.

TABLE 1. Positioning against representative prior work. ✓: addressed; $\sim$: partial; –: not addressed. PF: power flow.

Representative work	Distributed ADMM/consensus	AC PF validation	Comms-failure model	ML imputation of missing agents	Uncertainty quantification
Distributed AC-OPF [2], [7]	✓	✓	–	–	–
ADMM AC-OPF DER coord. [8]	✓	✓	–	–	–
Consensus / sharing DR [4], [6]	✓	–	–	–	–
Chance-constrained OPF [12], [13]	–	✓	–	–	✓
ADMM-OPF under delay [16]	✓	✓	✓	–	$\sim$
Resilient distributed opt. [19]	✓	–	✓	–	–
Learning-augmented ADMM-OPF [20]	✓	✓	–	$\sim$	–
Distributed TCL control [34], [35]	✓	–	$\sim$	–	–
ML telemetry imputation [21], [23]	–	–	–	✓	–
Thermal-DR flexibility [27], [28]	–	–	–	–	–
This work	✓	✓	✓	✓	✓

IV. METHODOLOGY

This section states the method in general terms; the concrete feeder, agent population, weights, and tariff used to instantiate it are given in Section V.

A. SHARING-ADMM COORDINATION

Let $x_i \in \mathbb{R}^T$ be the heating schedule of home i, h_i its baseline, and $B \in \mathbb{R}^T$ the fixed background aggregate. The coordinator flattens the total feeder load toward a constant target c while respecting each home's comfort. We solve the sharing problem

$$\min_{\{x_i\}} \sum_{i=1}^{N} \frac{\lambda}{2} \|x_i - h_i\|_2^2 + \frac{\mu}{2} \left\| \sum_{i=1}^{N} x_i + B - c\,\mathbf{1} \right\|_2^2 \quad (2)$$

subject to (1) for every i, where c is the constant level that conserves total daily energy, the weight λ penalizes comfort deviation, the weight μ penalizes departure of the total load from a flat profile, and $\mathbf{1}$ is the all-ones vector. The first term keeps each home close to its comfortable baseline; the second rewards a flat aggregate. Because daily energy is conserved, c is a constant and the objective is exactly the variance of the total load — the scheme is valley-filling, not curtailment.

Following the sharing form of ADMM [1], we introduce an auxiliary aggregate variable z and a scaled dual u with penalty parameter ρ_{ADMM}, and iterate three steps. The **x-update** solves each home's local problem in parallel: a comfort-plus-penalty proximal step followed by projection of the result onto the box-and-energy-plane feasible set of (1). The **z-update** is the closed-form average that drives the shared aggregate toward the flat target. The **u-update** is the standard scaled dual ascent on the sharing residual. Algorithm 1 summarizes one full sweep.

The x-update projection is what makes the scheme physically meaningful: it guarantees that every iterate respects comfort and conserves daily energy, so the coordinated schedule is feasible for the homes by construction. The number of iterations to convergence is recorded as a tractability metric.

Algorithm 1: Sharing-ADMM heating coordination

Input: baselines $\{h_i\}$, background B, target c, weights λ, μ, penalty ρ_{ADMM}
Output: coordinated schedules $\{x_i\}$
initialize $x_i \leftarrow h_i$, $z \leftarrow 0$, $u \leftarrow 0$;
repeat
  foreach *home i (in parallel)* **do**
    proximal step on $\frac{\lambda}{2}\|x_i - h_i\|^2$ plus penalty toward $z - u$;
    project x_i onto the box-and-energy set of (1);
  $z \leftarrow$ closed-form average enforcing the flat aggregate target;
  $u \leftarrow u + \big(\sum_i x_i + B\big) - z$;
until *aggregate residual converged*;
return $\{x_i\}$;

1) Capped-energy projection in the x-update

We make the local step explicit because its structure is what keeps every iterate feasible. For home i the unconstrained comfort-plus-penalty objective is a separable strictly convex quadratic, so its minimizer is available in closed form. Writing $a_i = z - u$ for the current penalty anchor, the prox optimum is the per-step convex combination

$$\tilde{x}_i = \frac{\lambda\, h_i + \rho_{\mathrm{ADMM}}\, a_i}{\lambda + \rho_{\mathrm{ADMM}}}, \quad (3)$$

which trades the comfort pull toward h_i against the consensus pull toward a_i. This unconstrained point need not respect (1), so we Euclidean-project it onto the intersection of the comfort box $[(1-\alpha)h_{i,t}, (1+\alpha)h_{i,t}]$ and the daily-energy hyperplane $\sum_t x_{i,t} = \sum_t h_{i,t}$:

$$x_i = \arg\min_x \tfrac{1}{2}\|x - \tilde{x}_i\|_2^2 \quad \text{s.t.} \quad x \in \mathrm{box}_i, \quad \mathbf{1}^\top x = \mathbf{1}^\top h_i. \quad (4)$$

Introducing a single scalar dual ν_i for the energy constraint, the projection has the water-filling form $x_{i,t} = \mathrm{clip}_{[(1-\alpha)h_{i,t},\,(1+\alpha)h_{i,t}]}(\tilde{x}_{i,t} + \nu_i)$, and the value of ν_i that restores $\mathbf{1}^\top x_i = \mathbf{1}^\top h_i$ is found by a 1-D bisection on the scalar dual. Because the clipped sum is monotone non-decreasing in ν_i, the bisection is exact and costs only a handful of evaluations, so the whole x-update is closed-form up to a scalar root-find. The subsequent z-update is fully separable per time step — each component of z is the average of the corresponding aggregate-plus-dual component driven toward the flat target c — and therefore also closed form. Both updates are vectorized across agents, so a full sweep remains inexpensive as the home population grows.

B. IMPUTATION OF NON-RESPONSIVE AGENTS

When agent i is non-responsive its baseline h_i is unavailable to the coordinator, so we replace it with a learned estimate $\hat{h}_i$. We train a regressor that maps a small feature vector — ambient temperature, the sine and cosine of the hour of day, and the agent's nameplate heating level — to instantaneous heating power. The model is trained on the responsive population and applied to each silent agent, yielding $\hat{x}_{i,t}$ that the coordinator treats as the agent's contribution. The default estimator is a gradient-boosted-tree regressor [26]; several learned and naive alternatives are compared in Section VI. Generalization to unseen homes is assessed by leave-agents-out cross-validation, which holds out entire agents rather than random samples and therefore measures the realistic case of imputing a home the model never saw.

C. POWER-FLOW VALIDATION LOOP

The novel stage closes the loop on physics. For each scenario (uncoordinated, ideal, and each non-responsive fraction ϱ) we inject the resulting aggregate into a radial distribution-feeder model with an explicit distribution transformer, distributing the demand across the feeder's load points, and solve an AC power flow at every time step. Per scenario we record the worst-of-day maximum transformer loading and minimum bus voltage. A scenario is flagged as violating if the transformer exceeds its thermal rating at any step. This turns an abstract kilowatt signal into a binary feeder verdict and a pair of continuous severity measures.

Two modeling choices are methodological rather than case-specific. Sizing the binding constraint into overload is what makes the experiment decision-relevant: a feeder comfortably within limits in the uncoordinated case has nothing to gain from demand response, so the transformer is rated such that the uncoordinated winter peak is in overload, placing the binding constraint exactly where coordination must act to clear it. Solving every step rather than only the aggregate-peak step matters because imputation error can shift the worst instant: the uncoordinated worst is the evening peak, but under heavy imputation the binding step can migrate, and an aggregate-peak-only check would miss it. Reporting the worst-of-day envelope across the whole horizon therefore gives a conservative, instant-agnostic verdict.

D. UNCERTAINTY QUANTIFICATION

The deterministic sweep fixes one failing-agent subset and the raw point forecast per fraction. To quantify risk we run a Monte-Carlo analysis over two sources of uncertainty: which agents fall silent (drawn at random) and the imputation error (the silent agents' estimates are perturbed by Gaussian residuals matched to the imputer's cross-validated error). For each fraction we draw many such realizations, re-solve the coordination, and record the distribution of the daily peak. Because the worst-of-day network metrics are monotone functions of that peak — the transformer sees the whole aggregate regardless of spatial placement — each realization's peak is mapped through a precomputed peak-to-loading curve, making the Monte-Carlo sweep inexpensive while reproducing the deterministic operating points exactly. We report the mean, a central band, and the probability that the transformer is in violation.

TABLE 2. Principal case-study parameters.

Parameter	Value
Homes N	74
Horizon / resolution	96 steps / 15 min
Coldest-day min. temperature	$-21.9\,°\mathrm{C}$
Comfort band α	0.5
Comfort weight λ	0.15
Flattening weight μ	1.0
ADMM penalty ρ_{ADMM}	1.0
Non-responsive fraction ϱ	$\{0.0, 0.2, 0.4, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.9, 1.0\}$
TOU off / mid / peak (CAD/kWh)	0.078 / 0.128 / 0.158
Feeder	Synthetic LV residential, 0.4 kV
Transformer	400 kVA MV/LV, 4 parallel LV feeders
Power factor	0.95
Uncoordinated transformer loading	109.6%
Power flow	AC Newton-Raphson
Software	pandapower 3.4.0, scikit-learn 1.6.1, LightGB

V. CASE STUDY SETUP

The study runs on a single synthetic Trois-Rivières cold day with a minimum ambient temperature of $-21.9\,°\mathrm{C}$, discretized at 15-minute resolution into $T = 96$ steps. We coordinate $N = 74$ electric-heating homes with comfort band $\alpha = 0.5$, comfort weight $\lambda = 0.15$, flattening weight $\mu = 1.0$, and ADMM penalty $\rho_{\mathrm{ADMM}} = 1.0$. The communication-failure fraction is swept over the twelve points $\varrho \in \{0.0, 0.2, 0.4, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.9, 1.0\}$.

Energy cost is evaluated against a time-of-use tariff with an off-peak rate of 0.078, a mid-peak rate of 0.128, and an evening-peak rate of 0.158 CAD/kWh. Physical feasibility uses a synthetic LV residential feeder of 74 literal homes behind a single 400 kVA MV/LV distribution transformer (25 kV / 0.4 kV) split across four parallel LV feeders, with all loads at a power factor of 0.95 and modeled one-for-one (no artificial scaling). The uncoordinated coincident peak loads the transformer to 109.6%. The software stack is pandapower 3.4.0 for power flow, scikit-learn 1.6.1 for the cross-validation harness, and LightGBM 4.6.0 for the imputer. Table 2 collects the principal parameters, and the reproducibility of the run is discussed below.

TABLE 3. Computational pipeline (principal stages).

#	Stage	Principal output
1	Weather generation	Synthetic cold-day temperature
2	Baseline heating	Per-home $h_{i,t}$ profiles
3	Background load	Fixed non-deferrable aggregate B
4	ADMM coordination	Ideal $x_{i,t}$, iteration count
5	Imputer training	Estimator + cross-validation metrics
6	Communication sweep	Per-ϱ imputed aggregates
7	Power-flow validation	$V_{\min}$, transformer loading, verdict
8	Uncertainty + comparison	Violation probabilities, method ranking

A. REPRODUCIBILITY

The study is designed to be reproduced exactly. It depends on no proprietary or field-measured data: the cold day, per-home thermal baselines, and background loads are all synthetic and generated from fixed random seeds, so the inputs are defined entirely by the parameters in Table 2 and regenerate identically on any machine. The computation is a single automated pipeline (Table 3) that runs end-to-end from that configuration without manual intervention, and it is built on open-source numerical libraries (pandapower for power flow, scikit-learn and LightGBM for the imputer), so no closed component stands between the configuration and the reported numbers. Each stage writes its outputs with an integrity digest, every figure and table in this paper is generated from those recorded outputs rather than transcribed by hand, and a regression check pins the headline quantities so that an accidental change is caught. The implementation and configuration are released with the paper. We claim re-runnable reproducibility on this configuration — not statistical generality across feeders or climates, which the limitations discuss.

VI. RESULTS

A. AGGREGATE COORDINATION

Figure 1 shows the cold-day aggregate load. The uncoordinated profile exhibits the classic double peak — a morning and an evening heating surge — whereas the ideal ADMM coordination flattens it to a nearly constant ≈ 362.3 kW. The $\varrho = 0.8$ trace, in which most agents are imputed, recovers much of the flattening but leaves residual peaks where imputation error accumulates.

Table 4 quantifies the aggregate key performance indicators. Ideal coordination cuts the peak from 401.6 to 362.3 kW, a 9.8% reduction, and drives the peak-to-average ratio (PAR) from 1.109 down to 1.000 — essentially a flat load. A PAR of 1.000 is the practical floor: it means the coordinated aggregate is constant to within a tenth of a percent across the entire day, so the flattening objective in (2) has been met almost exactly. Energy cost falls slightly, from 909.5 to 895.4 CAD — a 1.55% reduction — because the daily-energy constraint conserves total heating; the saving comes from shifting load out of the evening-peak TOU window (0.158 CAD/kWh) toward off-peak hours (0.078 CAD/kWh) rather than from using less energy. The

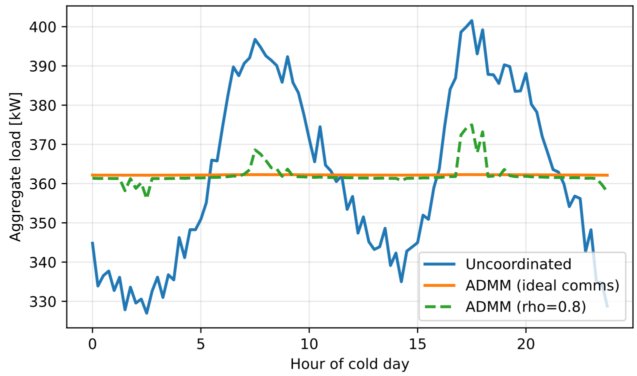

FIGURE 1. Aggregate cold-day load: uncoordinated double-peak versus the flat ADMM-ideal profile (≈ 362.3 kW) and the $\varrho = 0.8$ imputed case with residual peaks. Here ϱ is the non-responsive fraction.

modest size of the cost reduction is itself a useful result, and an honest one: the objective in (2) flattens the aggregate (it penalizes departure from a constant level), it does not chase the TOU price signal. A purely flat profile still places substantial load in the priced evening window, so the bill moves only as a side effect of peak-shaving. With energy conserved, demand response of this kind buys capacity relief (a lower peak) far more than it buys energy savings, and the case for it must rest on the network benefit rather than the bill.

1) Comfort preservation

The flattening is achieved without sacrificing thermal comfort. The mean absolute heating deviation of the ideal coordinated schedule from the thermostat baseline is 0.264 kW per home-step — small against the ≈ 5 kW heating level the homes operate at. This is comfort preserved by construction: the projection (4) keeps every step inside the $\pm 50\%$ band of its baseline and conserves each home's daily heating energy, so the coordinator can only re-time heat, never withhold it. The result is a feeder-level flat profile that is, at the level of each individual home, only a gentle nudge of when the thermostat draws power.

2) A plateau-and-ramp transition

Swept finely (Table 4), the degradation of coordination with the non-responsive fraction is not a smooth decay but a plateau followed by a ramp. From $\varrho = 0$ to $\varrho \approx 0.70$ the coordinated peak stays clamped at the ideal optimum (362.3–362.6 kW, PAR within 0.1% of unity); beyond a knee near $\varrho = 0.72$ it rises approximately linearly toward the uncoordinated peak, reaching 368.5, 375.0, 389.6, and finally 401.9 kW at $\varrho = 0.75, 0.8, 0.9$, and 1.0. This shape is the signature of a flexibility-budget mechanism, and it has two causes. First, an imputed home is pinned to its forecast — which reproduces its natural, peaky heating shape, not a flattened one — so it leaves the coordination pool; degradation is therefore driven mainly by the loss of controllable flexibility, not by raw forecast error (which

TABLE 4. Aggregate KPIs by non-responsive fraction ϱ (selected points; the full 12-point sweep appears in Figure 2). Iter.: ADMM iterations to converge.

Scenario	Peak (kW)	PAR	Cost (CAD)	RMSE (kW)	Ite
Uncoordinated	401.6	1.1090	909.5	—	-
Coordinated ideal	362.3	1.0000	895.4	—	
ϱ=0.4	362.4	1.0003	895.4	0.305	[illegible]
ϱ=0.55	362.4	1.0004	895.5	0.302	2
ϱ=0.6	362.4	1.0004	895.5	0.304	2
ϱ=0.65	362.5	1.0006	895.5	0.303	3
ϱ=0.7	362.6	1.0008	895.5	0.301	6
ϱ=0.75	368.5	1.0171	895.8	0.305	3
ϱ=0.8	375.0	1.0350	896.6	0.304	3
ϱ=0.9	389.6	1.0753	902.6	0.303	30
ϱ=1.0	401.9	1.1093	909.5	0.304	300

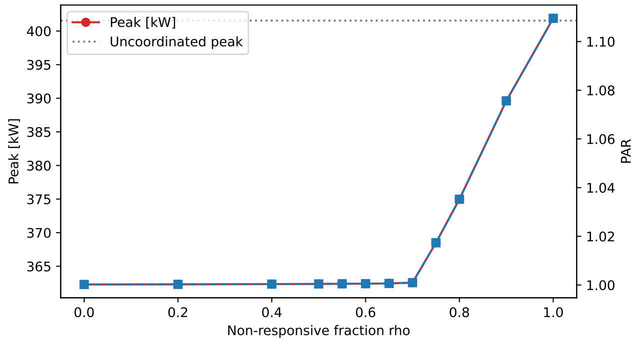

FIGURE 2. Aggregate peak and PAR versus the non-responsive fraction ϱ. The horizontal axis labelled "rho" in the plotted image denotes this fraction ϱ, not the ADMM penalty ρ_{ADMM}.

stays small, RMSE $\approx$ 0.3 kW, last column). Second, the remaining responsive homes flatten the total only up to their collective $\pm 50\%$ deferral budget: as long as that budget exceeds the peakiness injected by the pinned population, the peak is held exactly at the optimum (the plateau); once it is exhausted — around $\varrho \approx 0.72$, where only ~ 20 of 74 homes remain controllable — the residual peak passes through, and it does so linearly. The transition is thus sharp at its onset (a binding-constraint knee) but smooth thereafter (a linear ramp); the earlier impression of a sudden step was an artifact of coarse sampling between $\varrho = 0.7$ and 1.0.

B. IMPUTATION ACCURACY AND ROBUSTNESS

The LightGBM imputer is a competent forecaster of an unseen home's heating. Under 5-fold leave-agents-out cross-validation it achieves an RMSE of 0.304 kW and an MAE of 0.255 kW, against a heating standard deviation of 0.736 kW. At roughly 41% of the natural variability, the imputer is clearly better than predicting the mean and is accurate enough that the coordinated aggregate barely moves for non-responsive fractions below the flexibility knee, as seen in the per-scenario RMSE column of Table 4. Notably the per-scenario imputation RMSE is nearly flat in ϱ (≈ 0.30 kW) even as the aggregate degrades sharply — direct evidence that the damage past the knee comes from the loss of controllable agents, not from worsening forecasts.

Figure 2 plots peak and PAR against ϱ over the full 12-point sweep. Both remain clamped at the optimum through $\varrho \approx 0.70$ and then rise approximately linearly to the uncoordinated values at $\varrho = 1$ — a plateau-and-ramp, not a gradual decay, with the ramp onset (knee) marking exhaustion of the responsive deferral budget.

A secondary finding concerns tractability rather than outcome quality. The ADMM iteration count grows steeply with ϱ: 8 iterations for the ideal case, then 18, 27, 28, and 36 through $\varrho = 0.65$, rising to 65 at $\varrho = 0.70$ and then hitting the 300-iteration cap at $\varrho = 0.75$ and beyond (all runs below the cap converged). Communication failure thus degrades coordination tractability — the effort to reach consensus — well before it degrades the coordinated outcome, an early-warning signal an operator could monitor.

The shape of this curve is informative, and it shares its knee with the peak and the network results. Through $\varrho = 0.70$ the iteration count stays bounded (≤ 65); at $\varrho = 0.75$ it more than quadruples to the 300 cap even though the peak there has only just begun to move (368.5 kW). The knee of the iteration curve therefore precedes the visible outcome degradation: the solver feels the flexibility budget run out — the sharing residual becomes hard to drive to zero as the imputed aggregate diverges from the one that physically forms — one sweep step before the peak and the transformer loading break. Iteration count is thus a cheap, purely software-side proxy for the harder-to-observe physical-feasibility margin, visible before any meter or power-flow check, because both respond to the same underlying quantity: the mismatch injected by imputation.

C. IMPUTATION METHOD COMPARISON AND THE VALUE OF IMPUTATION

The results so far use a single imputer (LightGBM). Two questions remain: how much does imputation itself buy, and how much does the choice of method matter? To answer them fairly we adopt a stricter realized-load model that separates belief from reality: a silent home physically draws its true (uncontrolled, peaky) heating, while the coordinator sees only an estimate and flattens the responsive homes against it; the feeder then experiences the realized aggregate — responsive coordinated schedules plus the true silent load plus background. "No imputation" is the coordinator flattening the responsive subset blind to the silent homes. We compare four learned estimators (LightGBM, random forest, ridge regression, k-nearest neighbours), a naive "mean-level" baseline (a flat profile at each silent home's known nameplate level — correct energy, no shape), and the no-imputation case.

Table 5 reports both layers of metric. Intrinsically (5-fold leave-agents-out), ridge regression is the most accurate (RMSE 0.280 kW, $R^2 = 0.70$), slightly ahead of LightGBM (0.304 kW) and k-NN (0.308 kW) and the random forest (0.342 kW); with this small population and four smooth features the linear model is hard to beat. Notably the

TABLE 5. Imputation methods: intrinsic accuracy (leave-agents-out CV) versus realized feeder impact at $\varrho = 0.6$ (mean over 200 random silent subsets). Lower realized loading and violation probability are better.

Method	RMSE (kW)	R^2	Peak (kW)	Load. (%)	P_{viol}
Ridge	0.280	0.70	366.9	99.8	0.17
k-NN	0.308	0.67	368.1	100.1	0.61
LightGBM	0.304	0.67	370.4	100.8	0.92
Random forest	0.342	0.62	369.4	100.5	0.86
Mean-level (naive)	0.312	0.65	376.6	102.5	1.00
No imputation	—	—	389.0	106.0	1.00

naive mean-level baseline is close to the learned methods on RMSE (0.312 kW) — because each home's heating is dominated by its mean level, which that baseline gets exactly right — so intrinsic RMSE alone badly under-rates the importance of getting the timing right.

The realized-impact columns and Figure 3 tell the operational story, and here the violation probability P_{viol} is the sharpest discriminator. Without imputation the realized peak climbs to 389.0 kW (106% loading) at $\varrho = 0.6$ and the transformer violates in every draw ($P_{\mathrm{viol}} = 1.00$), because the outcome depends entirely on which uncontrolled homes fall silent. Any learned imputer holds the realized peak near 367–370 kW ($\approx$ 100–101%): imputation buys roughly a 19 kW reduction (389 to 370 kW) and, more importantly, drops the violation probability from 1.00 (none) to as low as 0.17 (ridge). The naive mean-level baseline recovers part of the peak (376.6 kW) but still violates in every draw — direct evidence that reconstructing the shape, not merely the energy, is what lets the responsive homes counter the silent peaks. Among the learned methods P_{viol} now discriminates cleanly and tracks intrinsic forecast quality: ridge 0.17, k-NN 0.61, random forest 0.86, and LightGBM 0.92, with the realized-loading ranking none (106%) > naive (102.5%) > the ML cluster ($\approx$ 100–101%) > ideal (98.5%). The practical lesson is therefore twofold: using an imputer matters far more than which one — one versus none is the gap that dominates — and a simple ridge model is the best, cheapest choice on data of this size, while LightGBM remains a robust default that scales to the larger populations a utility would face.

D. NETWORK VALIDATION

The closed-loop contribution is summarized in Figure 4 and Table 6. The uncoordinated aggregate loads the 400 kVA transformer to 109.6% — a thermal violation — while the worst minimum voltage sits at a healthy 0.962 p.u. Ideal coordination clears the thermal overload: maximum transformer loading falls to 98.5% (no violation) and the worst minimum voltage stays healthy at 0.966 p.u. This is the core result: flattening the aggregate is not merely a statistical improvement, it restores feeder feasibility.

The cleared state is resilient to communication failure through $\varrho = 0.70$, where the transformer holds at 98.6% loading and does not violate. The deterministic crossing

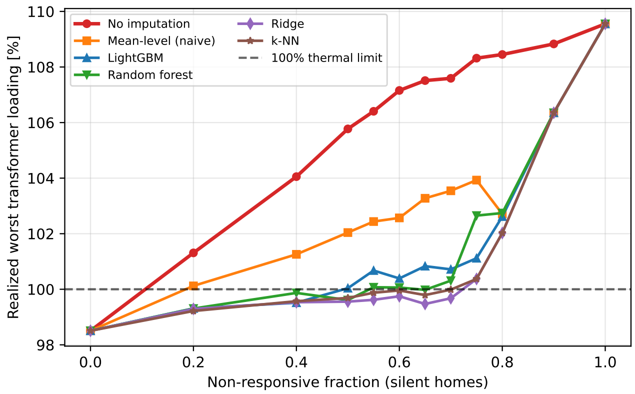

FIGURE 3. Realized worst transformer loading versus the non-responsive fraction for each imputation method under the realized-load model. No imputation (red) diverges early; the learned methods cluster near the 100% limit; the naive mean-level baseline sits between them. All converge to the uncoordinated 109.6% once every home is silent.

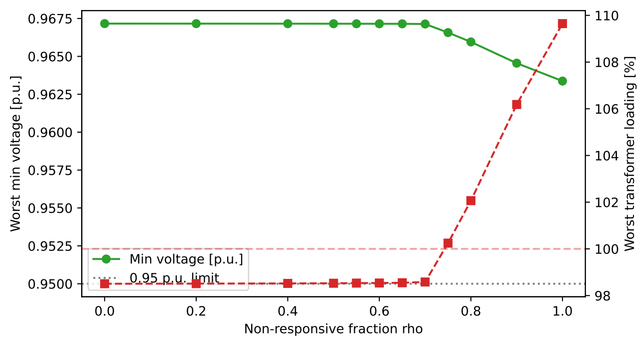

FIGURE 4. Worst minimum bus voltage (left axis) and worst maximum transformer loading (right axis) versus the non-responsive fraction ϱ — the headline feasibility figure. Loading stays cleared below 100% through $\varrho = 0.70$, crosses the limit at $\varrho = 0.75$, and ramps linearly to the uncoordinated 109.6% at $\varrho = 1$. The image's "rho" axis is the non-responsive fraction ϱ.

falls between $\varrho = 0.70$ (98.6%, cleared) and $\varrho = 0.75$ (100.2%, violated); beyond it the loading ramps linearly — 102.1%, 106.2% — back to the uncoordinated 109.6% once every home is imputed at $\varrho = 1$. The story is therefore 109.6% $\to$ 98.5% $\to$ 109.6%: coordination clears the overload, the clearance survives substantial non-responsiveness, and then the protection unwinds linearly as the controllable population vanishes. This pins the critical non-responsive fraction — beyond which imputation can no longer protect the transformer — to a narrow band around $\varrho = 0.72$.

a: Voltage stays healthy.

Across every scenario the worst minimum LV bus voltage stays in a comfortable 0.962–0.966 p.u. band and never approaches the 0.95 p.u. limit, so the feeder is voltage-comfortable throughout and the transformer thermal loading is the unambiguous binding constraint — a cleaner story than a benchmark that is stressed on voltage to begin with. The 100% transformer rating gives an unambiguous binary verdict, and coordination is what moves the feeder across it

TABLE 6. Network feasibility on the LV feeder (worst of day), selected fractions. Loading is the 400 kVA transformer's worst-of-day loading.

Scenario	Peak (kW)	$V_{\min}$ (p.u.)	Loading (%)	Violation
Uncoordinated	401.6	0.962	109.6	yes
Coordinated ideal	362.3	0.966	98.5	no
ϱ=0.4	362.4	0.966	98.5	no
ϱ=0.55	362.4	0.966	98.5	no
ϱ=0.7	362.6	0.966	98.6	no
ϱ=0.75	368.5	0.966	100.2	yes
ϱ=0.8	375.0	0.965	102.1	yes
ϱ=0.9	389.6	0.964	106.2	yes
ϱ=1.0	401.9	0.962	109.6	yes

TABLE 7. Monte-Carlo transformer loading and violation probability by fraction (200 draws each; mean [P5, P95]).

ϱ	Loading (%) [P5, P95]	P(violation)
0.40	98.5 [98.5, 98.5]	0.00
0.55	98.5 [98.5, 98.5]	0.00
0.70	99.1 [98.6, 99.95]	0.04
0.75	101.0 [100.1, 102.0]	0.96
0.80	102.5 [101.6, 103.4]	1.00
1.00	110.0 [109.2, 110.9]	1.00

from infeasible to feasible.

E. FORECAST UNCERTAINTY QUANTIFICATION

The network verdict above rests on a single realization per non-responsive fraction. To test whether that point estimate understates risk, we run the Monte-Carlo analysis of Section IV (200 draws per fraction over which agents fall silent and Gaussian forecast residuals of standard deviation 0.304 kW, the imputer's cross-validated RMSE), mapping each draw's daily peak to the worst-of-day transformer loading.

Figure 5 and Table 7 report the result (mean with [P5, P95] band). The probability of a thermal violation stays at 0 through $\varrho = 0.65$, ticks to 0.04 at $\varrho = 0.70$, jumps to 0.96 at $\varrho = 0.75$, and is 1.00 for $\varrho \geq 0.80$. The single-realization analysis reported $\varrho = 0.70$ as comfortably compliant (98.6% loading), but once the uncertainty in which agents fail and in the forecast is accounted for, the transformer still clears at $\varrho = 0.70$ (only 4% of draws breach) yet exceeds its limit in 96% of draws by $\varrho = 0.75$: the deterministic estimate understated how abruptly the risk rises. The [P5, P95] band is tight on the plateau — forecast errors average out across a large responsive population — and widens at the knee ($[98.6, 99.95]\%$ at $\varrho = 0.70$, $[100.1, 102.0]\%$ at $\varrho = 0.75$) before riding the ramp upward. This reframes the operator takeaway: the critical transition is not a single deterministic fraction but a probabilistic risk threshold centered near $\varrho = 0.72$, where the breach probability swings from near zero to near certainty across a single sweep step.

F. DISCUSSION

Three observations tie the results together. First, the aggregate, tractability, and feasibility curves all share the same knee near $\varrho = 0.72$: the iteration count, the peak, and the worst transformer loading all stay benign on the plateau through $\varrho \approx 0.70$, break at the same knee, and ramp together thereafter. This suggests that a single observable quantity — the divergence between the imputed and the realized aggregate, governed by the responsive flexibility budget — drives all three, and that an operator monitoring ADMM convergence effort has an early proxy for impending feeder violation (the iteration knee leads the loading knee by one sweep step). Second, the network result is not a restatement of the aggregate result: a 9.8% peak cut is a statistical improvement, but clearing a 109.6% transformer overload to 98.5% is an operational state change from infeasible to feasible, and the two need not coincide on a different feeder or load allocation. Third, the unwinding past the knee is graceful rather than catastrophic — the loading ramps linearly to 109.6% and the power flow always converges, never collapsing — which indicates that the imputer keeps the system in a recoverable regime even when most agents are silent. Together these support the central claim that coordination quality should be measured by feeder feasibility, and that imputation extends the range of communication failure over which that feasibility holds — here, up to a critical non-responsive fraction near $\varrho = 0.72$.

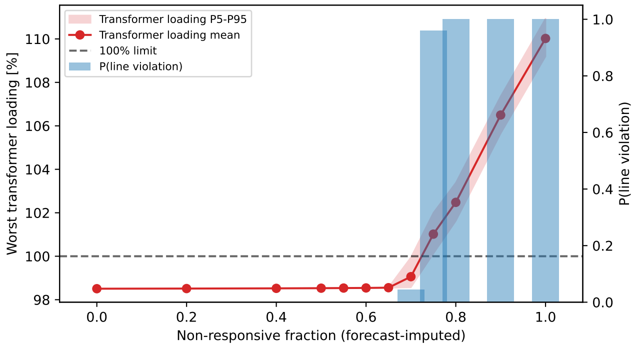

FIGURE 5. Monte-Carlo forecast-uncertainty sweep (200 draws per fraction). Worst maximum transformer loading mean with its P5–P95 band versus the non-responsive fraction ϱ (left axis, with the 100% thermal limit), and the probability of a transformer violation as bars (right axis). The image's "rho" axis is the non-responsive fraction ϱ.

VII. CONCLUSION

We presented a network-validated pipeline that coordinates cold-climate electric heating with sharing-ADMM, reconstructs non-responsive agents with a LightGBM imputer, and — the closing contribution — validates the coordinated aggregate on a synthetic LV residential feeder of 74 all-electric homes behind a 400 kVA distribution transformer with AC power flow. The closed loop turns abstract flattening metrics into a feeder verdict: ideal coordination cuts the aggregate peak by 9.8% and clears a 109.6% transformer thermal overload down to 98.5%, while LV voltage stays healthy (≈ 0.96 p.u., never approaching the 0.95 p.u. limit), so the transformer is the unambiguous binding constraint.

The cleared state is resilient to communication failure through a non-responsive fraction of about 0.70, and a finely-resolved sweep shows the protection then unwinds not as a sudden step but as a plateau followed by a roughly linear ramp: the worst transformer loading stays clamped near 98.5% up to $\varrho \approx 0.70$, crosses the 100% limit at $\varrho = 0.75$, and rises linearly to the uncoordinated 109.6% as the last homes fall silent. A Monte-Carlo analysis over which agents fail and the forecast error reframes this critical fraction as a probabilistic risk threshold rather than a single deterministic point: the probability of a thermal violation stays at 0 through $\varrho = 0.65$, ticks to 0.04 at $\varrho = 0.70$, and reaches 0.96 by $\varrho = 0.75$, so the deterministic compliance at $\varrho = 0.70$ understated the true risk. A secondary finding is that the ADMM iteration count explodes at the same knee (from 65 to the 300-iteration cap between $\varrho = 0.70$ and 0.75) one sweep step before the outcome degrades, offering an early-warning signal.

The study has clear limitations. It covers a single cold day and a single feeder, and the binding constraint is the distribution transformer's thermal loading; the communication-failure model is a stochastic non-responsive fraction rather than a protocol-level simulation of delays, drops, and retransmissions. Future work will extend the loop to multiple feeders and days, study the spatial placement of agents across the LV feeders (which determines how much each contributes to the binding constraint), and couple the coordinator to a flexibility market so that the cleared headroom can be priced and settled.

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